Supplementary Material: Controlling Evidence for Interpretable Artificial Intelligence-Assisted Gait Assessment

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This supplementary file contains methodological audits, additional retrieval sensitivity results, model-specific results, and prompt templates that support the main manuscript.

Supplementary Tables

Table 1: Supplementary Robustness and methodological audits. These audits address baseline strength, TGR confounding, context budget, parse/schema reliability, prompt sensitivity, modern flat baselines, and semantic usefulness.

Audit	Result	Interpretation
Generation/schema status	2,460 total records; 711 no-feature records; 1,749 attempted LLM generations; 0 unrecoverable parser or schema failures	No generated record was excluded because of malformed output, but recoverable JSON fallback extractions were not logged separately
Context budget	Overall FR mean = 3.187 contexts; TGR mean = 2.765 nodes; TGR used fewer contexts in 742 records, equal contexts in 843, and more contexts in 164	TGR did not have a larger context budget overall; the amputee dataset was nearly equal (FR 2.557, TGR 2.602)
Deterministic taxonomy selector parity	Final TGR contexts equaled deterministic exact-tag taxonomy selection in 264/264 amputee records and 219/219 GAITEX records; equality was lower in Clinical Gait Signals (80/540) and Health&Gait (0/726)	The strongest positive TGR result was not driven by unconstrained LLM node selection; weaker datasets were more sensitive to taxonomy routing and selector behavior
Unsupported claims	Overall unsupported claims were 115 for FR and 234 for TGR; in the amputee dataset, unsupported claims decreased from 19 with FR to 0 with TGR; in Health&Gait they increased from 11 to 174	TGR is not globally safer; it helps when taxonomy structure matches the evidence and can hurt when it does not
Prompt-sensitivity smoke	16 additional generations using two prompt variants, two models, two methods, and two cases. In the Health&Gait case, both methods had 0 unsupported claims. In the amputee case, FR produced 4 unsupported claims and TGR produced 0	This small smoke test supports the amputee direction under prompt perturbation, but it is not a full prompt robustness study
Targeted ablation and modern-flat baselines	72 additional generations on a stratified informative sample: flat coverage-matched retrieval, deterministic taxonomy without LLM node selection, BM25, dense Ollama retrieval, and hybrid feature-aware variants	Deterministic taxonomy and coverage-matched flat retrieval were nearly tied overall; pure BM25 and pure dense retrieval lost feature coverage, while feature-aware hybrid retrieval achieved the strongest automatic grounding in the sample
Modern retriever sensitivity analysis	216 additional records over a normalized 24-unit evidence corpus, comparing BM25, dense, biomedical dense, hybrid BM25+dense, and LLM-reranked hybrid retrieval across top-k 3, 5, and 8	Hybrid BM25+dense was strongest overall; LLM reranking, biomedical dense retrieval, and dense-only retrieval did not dominate this bounded check
Secondary semantic audit	4,517 blinded LLM-judge records for Clinical Gait Signals and Health&Gait. Mean usefulness was 3.447 for FR and 3.383 for TGR; winner counts were FR 1,955, TGR 1,775, tie 787	Semantic usefulness did not automatically improve with TGR, again showing that the best method depends on the evidence

Table 2: Supplementary Full Ontology-Guided Graph Retrieval (OGR) grounding metrics. OGR maps case features to GADO concepts, expands one ontology-graph hop, and retrieves the top 3 linked evidence units before using the same generator and support checker as FR and TGR.

Dataset	N	Contexts	Coverage	Faithfulness	Unsup. rate	Citation	Quality
Clinical Gait Signals	540	3.0	0.9463	0.9840	0.0160	1.0000	2.9143
GAITEX	219	3.0	0.9932	0.9989	0.0011	1.0000	2.9909
Health&Gait	726	3.0	0.9970	0.9979	0.0021	1.0000	2.9929
Above-Knee Amputee Gait	264	3.0	0.9575	0.9599	0.0401	1.0000	2.8773
Overall	1749	3.0	0.9749	0.9880	0.0120	1.0000	2.9509

Table 3: Supplementary Targeted ablation and modern flat-baseline results. Each method was evaluated on the same 12 case-model pairs: one informative case per dataset and generator model. Context counts were matched to deterministic taxonomy node count for each case.

Method	N	Mean contexts	Feature coverage	Faithfulness	Unsupported rate	Citation accuracy	Quality
Flat coverage-matched	12	2.5833	0.9578	0.9697	0.0303	1.0000	2.8972
Deterministic taxonomy	12	2.5833	0.9881	0.9881	0.0119	1.0000	2.9643
BM25 flat	12	2.5833	0.9246	0.9722	0.0278	1.0000	2.8691
Hybrid BM25+feature overlap	12	2.5833	1.0000	1.0000	0.0000	1.0000	3.0000
Dense Ollama flat	12	2.5833	0.3651	0.3989	0.6011	0.7500	0.9129
Hybrid dense+feature overlap	12	2.5833	0.9578	0.9697	0.0303	1.0000	2.8972

Table 4: Supplementary Modern retriever sensitivity benchmark on the stratified sanity subset. Higher coverage, faithfulness, citation accuracy, and quality are better; lower unsupported rate is better.

Method	N	Coverage	Faithfulness	Unsup. rate	Citation	Quality
Hybrid BM25+dense	36	0.9629	0.9889	0.0111	1.0000	2.9407
Hybrid + LLM rerank	36	0.9579	0.9639	0.0361	1.0000	2.8857
Dense, all-MiniLM-L12-v2	36	0.9205	0.9389	0.0611	0.9722	2.7705
BM25	36	0.9114	0.9288	0.0712	0.9722	2.7412
Dense, PubMedBERT-MS MARCO	36	0.8580	0.8841	0.1159	1.0000	2.6263
Dense, Ollama llama3.1:8b	36	0.8062	0.8417	0.1583	0.9722	2.4618

Table 5: Supplementary Model-specific composite quality scores by dataset and retrieval method. The best value in each dataset/model row is shown in the final column.

Dataset/model	FR quality	TGR quality	OGR quality	Best	Pattern
Clinical Gait Signals / `llama3.1:8b`	2.9094	2.8641	2.8384	FR	FR strongest
Clinical Gait Signals / `mistral-small:24b`	2.9573	2.8318	2.9649	OGR	OGR slightly strongest
Clinical Gait Signals / `qwen3:14b`	2.9504	2.9836	2.9398	TGR	TGR strongest
GAITEX / `llama3.1:8b`	2.8185	2.8523	2.9795	OGR	OGR strongest
GAITEX / `mistral-small:24b`	2.9726	2.9760	2.9932	OGR	OGR strongest
GAITEX / `qwen3:14b`	2.9760	3.0000	3.0000	TGR/OGR	Tied
Health&Gait / `llama3.1:8b`	2.9891	1.5501	2.9787	FR	FR strongest
Health&Gait / `mistral-small:24b`	2.9986	2.8223	3.0000	OGR	OGR slightly strongest
Health&Gait / `qwen3:14b`	2.9824	3.0000	3.0000	TGR/OGR	Tied
Amputee / `llama3.1:8b`	2.7981	3.0000	2.8739	TGR	TGR strongest
Amputee / `mistral-small:24b`	2.9913	3.0000	2.8761	TGR	TGR strongest
Amputee / `qwen3:14b`	2.9739	3.0000	2.8818	TGR	TGR strongest